

Phase Hunter: mapping the ANNNI phase diagram under noise

Q-SITE 2026 Open Challenge · Scientific Track · Tanish Singh Rajpal (Carnegie Mellon University, INI)

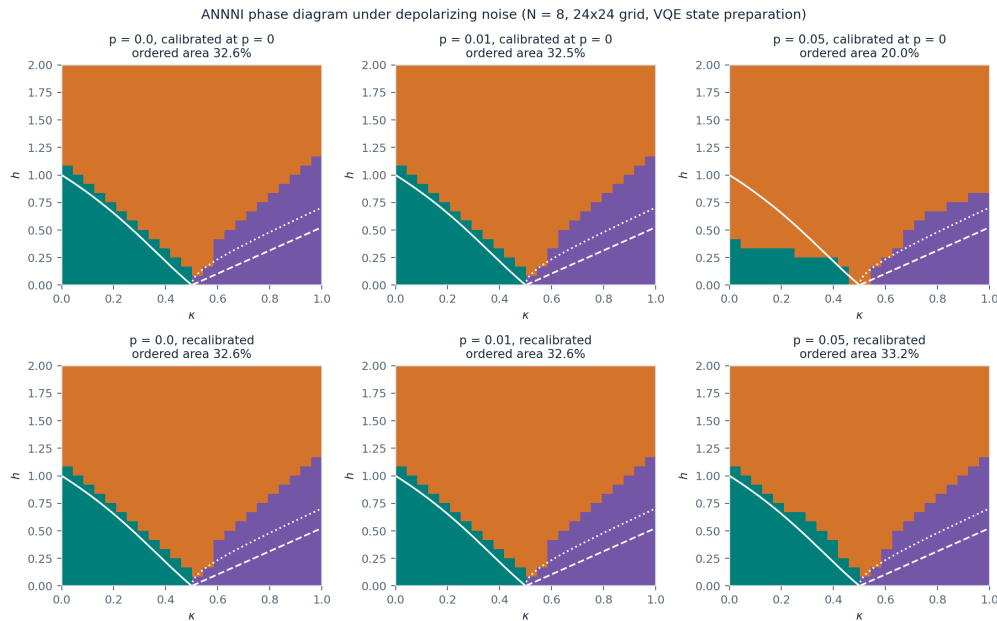
Method

We study the 1D ANNNI chain, $H = -\sum_i Z_i Z_{i+1} + \kappa \sum_i Z_i Z_{i+2} - h \sum_i X_i$, on a ring, using two observables an experiment can actually collect: the nearest-neighbour correlator $\langle Z_i Z_{i+1} \rangle$ and the next-nearest one $\langle Z_i Z_{i+2} \rangle$. Phases are assigned by unsupervised clustering of that pair, with the three cluster centres seeded at corners of the plane whose phase is not in doubt. The classifier never sees the analytic transition lines, so agreement with them is a real test rather than a fit.

For the noisy runs each ground state is prepared by a hardware-efficient variational circuit fitted on **default.qubit**, and the same circuit is then executed on **default.mixed** with a depolarizing channel after every CNOT. Every fit is checked against exact diagonalisation and refitted from a cold start when it lands above tolerance: 78% of points needed that, and the final mean energy error is 0.149 (1.5% of $|E|$), with the prepared states reproducing the exact correlators to 0.027 and 0.042. The scan covers a 24×24 grid at $N = 8$ and took 58 minutes.

Result 1 — the clean diagram

Exact ground states on a 40×40 grid at $N = 8$ reproduce the four-phase structure with **92.3%** of cells matching the analytic labels once the floating band is excluded (88.9% counting it as error). The residual disagreement is not scattered: it is a band just above the analytic lines, where a finite ring still holds order that the thermodynamic-limit formulas say has gone.



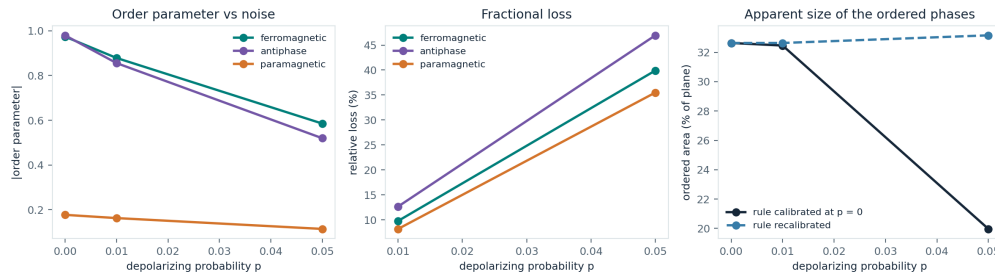
Phase diagrams at $p = 0, 0.01, 0.05$. Top: classified with a rule calibrated on the clean data. Bottom: the rule re-fitted on each noisy dataset. White lines are the analytic boundaries.

Result 2 — noise moves the scale, not the boundary

Rule	$p = 0$	$p = 0.01$	$p = 0.05$
Ordered area, calibrated at $p = 0$	32.6%	32.5%	20.0%
Ordered area, recalibrated	32.6%	32.6%	33.2%
Agreement with analytic lines (recal.)	92.2%	92.2%	91.7%

Read with a fixed rule the ordered phases lose 39% of their area by $p = 0.05$; recalibrated they do not move at all. A depolarizing channel attenuates every correlator by roughly the same factor, so it destroys the *scale* of an order parameter while leaving the *location* of the transition intact. What noise really costs is signal-to-noise, and therefore shots — not the physics you are trying to locate.

Which phase suffers most is then a question about correlator range. Deep inside each region the fractional loss at $p = 0.05$ is **47% for the antiphase** against 40% for the ferromagnet and 35% for the paramagnet. The antiphase order parameter lives on $\langle Z_i Z_{i+2} \rangle$ — a longer-range object, with more gates between the two spins being compared — so the same per-gate error costs it more. The handout hypothesises this; the numbers above measure it.



Left: order parameters versus p . Centre: fractional loss, where the antiphase separates from the ferromagnet. Right: apparent size of the ordered phases under a fixed versus a recalibrated rule.

Result 3 — a finite-size trap worth reporting

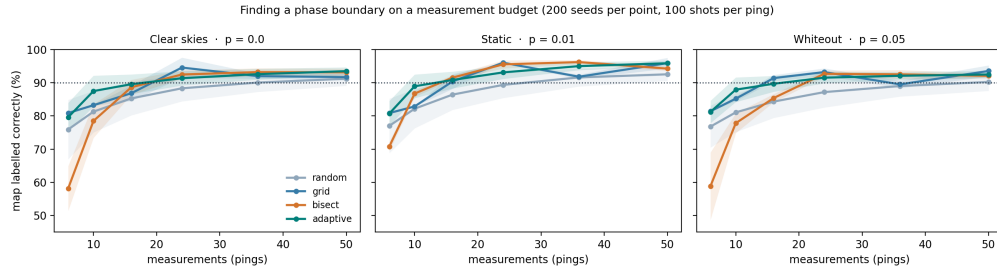
The antiphase is a period-4 pattern, so on a *ring* it only fits when the ring length is divisible by four. Our first full noisy scan ran at $N = 6$, where the pattern is geometrically frustrated out of existence: deep in the antiphase $\langle Z_i Z_{i+2} \rangle$ reaches only -0.33 instead of -0.99 , and agreement with the analytic lines collapses to 43.9% against 91.9% at $N = 8$ and 95.1% at $N = 12$. An hour of compute was lost to it. Anyone extending this work should use $N = 8$ or 12 and never 6 or 10; we kept $N = 6$ as a playable stage precisely because the failure is so visible.

Result 4 — how many measurements a boundary costs

Beyond the required deliverables we asked our own question: under a fixed budget of measurements, where should you spend them? Four strategies choose points, each ping returns the correlators with real shot noise at 100 shots, a boundary is reconstructed from those pings alone, and the reconstruction is scored against ground truth over 40 seeds per point.

Pings needed to label 90% of the map correctly	$p = 0$	$p = 0.01$	$p = 0.05$
Random sampling	36	36	50
Even grid	24	16	16
Per-column bisection	24	16	24
Adaptive (coarse sweep, then sample the edge)	24	16	24

Adaptive sampling pays where the budget is scarce — at 10 pings on the clean stage it labels 87.5% of the map correctly against 78.5% for bisection and 81.3% for random — and random sampling needs roughly twice the budget of any structured strategy to clear 90%. By 36–50 pings the structured strategies sit between 92% and 93% on the clean stage while random still trails at 90–91%, so the advantage is in the cheap regime rather than asymptotically, and that ceiling is our reconstruction and grid resolution rather than physics. Averaged over 200 seeds per point. The budget ladder is coarse, so a crossing whose mean sits within a standard error of the 90% line can move one rung on another machine; the script flags those itself, and 3 of the twelve crossings currently qualify. We report this as measured on our own pipeline, not as a general claim.

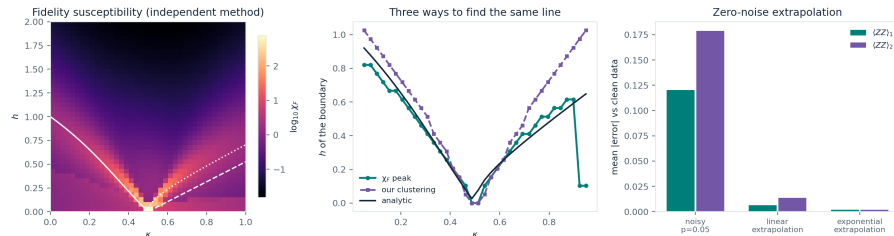


Accuracy versus measurement budget at each noise level; bands are one standard deviation over 40 seeds.

Result 5 — a second method, and getting the scale back

Two checks close the loop. The first is an independent method. The fidelity susceptibility, built from the overlap of ground states at neighbouring field values, shares no machinery with the correlator clustering: no order parameter, no centroids, no labels. Because the ordered phases of a finite ring are quasi-degenerate we compare whole ground manifolds rather than single eigenvectors, making the measure invariant to the arbitrary basis a degenerate solver returns. Its peak lands on the analytic line within 0.035 in h for $\kappa \leq 0.85$, below the grid spacing of 0.051 — and it fails where everything else here fails, rising to 0.28 in the floating corner. Two unrelated methods agreeing on the clean physics and losing the thread in the same place is a stronger statement than either alone.

The second is error mitigation, which turns Result 2 from a diagnosis into a repair. Holding runs at two noise levels, we extrapolate each correlator back to $p = 0$ and check it against clean data the extrapolation never saw. Richardson (linear in p) cuts the mean error on $\langle ZZ \rangle_1$ from 0.120 to 0.0065 and on $\langle ZZ \rangle_2$ from 0.179 to 0.0137; a two-point exponential fit, the right shape if depolarizing noise multiplies each correlator by a constant factor, reaches 0.0019 and 0.0020. Under the fixed $p = 0$ rule the mitigated data recovers 32.6% ordered area against 20.0% unmitigated: the 39% that noise erased comes back. The caveat is ours to state — our noise really is a depolarizing channel of known strength, the friendliest case extrapolation ever sees. It confirms the attenuation picture behind Result 2; it is not a claim about hardware.



Fidelity susceptibility, computed without any classifier; the boundary located three independent ways; extrapolation error against clean data.

The deliverable people can play

The same data drives a browser game: the phase diagram starts hidden, a player spends a budget of measurements, draws the boundary, and is scored against the stage's own ground truth. Its five stages are the physics above — 8 spins clean, at $p = 0.01$ and $p = 0.05$, 6 spins where the stripes cannot fit, and 12 spins where the borders sharpen — and each round crops a different window so the answer cannot be memorised. The opponent plays the bisection strategy of Result 4.

Limitations

$N = 8$ for the noisy scan and 12 for the largest clean stage; finite-size effects are visible as order surviving above the thermodynamic-limit boundaries, which caps our agreement near 92%. The floating phase is not resolved and is excluded from scoring. The variational ansatz is imperfect, which is why the headline result is a ratio between noise levels rather than absolute correlator values. One noise model and one circuit family were tested.

References and reproduction

Quantum Coalition, QSITE 2026 Scientific Track handout and starter kit (Hamiltonian convention, analytic Ising/BKT/KT lines, noise model, rubric). PennyLane demo *Phase transitions of the ANNNI model*; PennyLane challenge *A Noisy Heisenberg Model*; **default.mixed** and **DepolarizingChannel** documentation. P. Bak and J. von Boehm, Phys. Rev. B 21, 5297 (1980). Every number above renders from result JSON written by **scripts/**.